Isoliner · Topography: from DEM to catchments

A cheat sheet for the ”2. Topography” group · QGIS 3.16+ · Isoliner plugin · no GRASS or SAGA

Pipeline: from a DEM to Topo2Raster

Step 1. DEM - 2.01 Download DEM by extent. Set the extent, keep the defaults. Output: a 30 m terrain raster in a metric CRS.

Step 2. Contours - 1.04 Isolines from raster. Input: the DEM of step 1. Isoline step 10 m, value field name **ELEV**. Leave the minimum line length at 0, because short closed lines are hilltops. Turn contour polygons off.

Step 3. Streamlines - 6.02 River network. Input: the DEM of step 1. Threshold 1000 cells, the lower the threshold, the denser the network. Line vertices already run downstream, as Topo2Raster requires.

Step 4. Peak and pit elevations - 2.06 Peaks and pits. Input: the DEM of step 1. Window radius 500 m, drop 20 m. Output: points with the **z**, **drop** and **kind** (peak or pit) fields, the elevation also sits in the geometry Z. Without these points Topo2Raster cuts hilltops and hollows inside closed contours into flat pads.

Step 5. Water bodies and cliffs - 2.02 Download base topography by extent. The same extent. Take from OSM what a raster cannot give, that is water bodies and cliffs. OSM watercourses and peaks are a fallback for steps 3-4 when the ele field is filled.

Step 6. Terrain - 2.03 Topo2Raster (terrain from vectors). Contours from step 2 (the ELEV field), streamlines from step 3, elevation points from step 4 (the z field), lakes from step 5, the cliffs of step 5 into **Cliffs (smoothing barriers)** so the drop is not smeared. Cell 30 m, extent empty. Output: the terrain raster.

Check: run 1.04 over the result with the same step and overlay it on the contours of step 2. The lines must coincide. Contours from a digitized plan replace steps 1-2. All outputs land in the **Topography** group of the layer tree.

Tool reference: input, output, key parameters

Tool	Input	Output	Key parameters
2.01 Download DEM by extent	map extent	GeoTIFF float32, meters	source: GLO-30 (DSM), GEDTM30 (DTM, forest removed) or Mapzen (SRTM, ArcticDEM). Cell 30 m. CRS: empty = project/UTM
2.02 Download base topography by extent	map extent	watercourses, water bodies, peaks (ele), cliffs	coastline off. Large extent: shrink it or raise the extent area limit
1.04 Isolines from raster	DEM	lines with ELEV	step 5-10 m. Min. line length 0, short closed lines are hilltops. Default smoothing. Polygons off
2.03 Topo2Raster	contours (elevation field), points, streamlines, cliffs, lakes	GeoTIFF float32	points or contours required. Streamlines run downstream: OSM and 6.02 fit as is. Edge: node Z (river slope) > field elevation (plane) > shore minimum
2.04 Terrain preparation	DEM	prepared DEM	FPDEMS smoothing (edge-preserving) and/or depression filling (epsilon 0.001 for D8). Both by checkboxes
6.01 Flow and accumulation (D8)	DEM	directions (ArcGIS codes, byte) + accumulation (cells, float32)	codes: E=1, SE=2, S=4, SW=8, W=16, NW=32, N=64, NE=128, sink=0
6.02 River network	DEM	lines: order (Strahler), acc_out, length_m	threshold = head catchment / cell area. 1000 at 30 m ≈ 0.9 km²
6.03 Basins and watersheds	DEM, pour points (empty = mouths)	polygons: basin, area_m2 (+ label raster)	without points, basins start at mouths by threshold. Snap radius 150 m
2.05 Slope and aspect	DEM	two rasters, degrees	Horn 3×3 as gdaldem. Aspect: downslope azimuth, flats = -1
2.06 Peaks and pits	DEM	points: z, drop, kind (PointZ)	both signs always. A 500 m window radius keeps one peak per window, a 20 m drop discards bumps and small hollows
6.04 Gauge point report	DEM, gauge points	gauge watersheds (polygons), HTML report	a point snaps to the accumulation maximum within 150 m
6.05 Catchment of a line or an outline	DEM, lines or polygons	catchment polygon, area	a polygon is an intake as a whole, the outline and everything inside. For a pit the catchment does not depend on flow directions in the hollow

Tool	Input	Output	Key parameters
2.12 Cut and fill	two surfaces, or a surface and an elevation	difference raster, HTML statement	the sign is “after minus before”. Grids are aligned bilinearly. The dead band discards background noise

Common mistakes

- **A degree CRS.** The group works in meters. 2.01 reprojects itself: an empty target CRS = the project CRS if metric, otherwise UTM at the center. Custom CRSs without an EPSG code work like any other.
- **The river threshold is tuned.** Start at 1000 and change it threefold in the needed direction. Check by overlaying the 6.02 network on the OSM watercourses.
- **The Overpass server limits requests.** It rejects an oversized OSM request, and the tool then switches to a mirror. Keep the extent modest.
- **Closed basins.** Depression filling erases karst and subsidence troughs, so uncheck it.

Data: Copernicus DEM © ESA · GEDTM30 © OpenGeoHub CC BY 4.0 · Mapzen Terrain Tiles (credits: github.com/tilezen/joerd) · © OpenStreetMap, ODbL. Plugin: plugins.qgis.org/plugins/grid_isolines · The manual ships with the plugin (doc/Isoliner_en.pdf) · Inform++ LLC · www.informpp.ru